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What are the factors that contribute to CO2 emissions differences across European countries in the private aviation sector?

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Table of contents

1 Introduction **3**
1.1 General Context . . . 3
1.2 Private Aviation . . . 3
1.3 Dataset . . . 4

2 IPAT and Kaya Decomposition **5**
2.1 What is an IPAT and a Kaya decomposition ? . . . 5
2.2 Kaya decomposition for private aviation . . . 6
2.3 Limitations . . . 6
2.4 Unobserved factors . . . 7

3 Analysis between European Countries **8**
3.1 Observations and Comparisons Between Countries . . . 10
3.2 Case study: France, UK and Germany . . . 11

4 City Analysis: London, Geneva, Paris, and Nice **12**
4.1 Observations and Comparisons Between Cities . . . 13
4.2 Correlation Analysis . . . 13

5 Different ways of meeting Europe Net-zero 2050 target **15**
5.1 Reducing IPAT ratios . . . 15
5.2 Other transport alternatives . . . 16
5.3 Carbon compensation . . . 17

6 Conclusion **18**

7 Appendix **21**

CHAPTER 1

INTRODUCTION

1.1 GENERAL CONTEXT

The EU aims to be carbon neutral by 2050[1] in order to become the first climate-neutral continent. These assertions are not merely empty talk, the European Climate Law enshrines this goal into the law. These ambitious claims are clearly at odds with private aviation. It is tempting to dismiss private jet emissions as they represent a fraction of the 3.8% of total EU CO2 emissions that aviation emits.[2] However, once one considers emissions per capita then we can see the importance of tackling this issue. Compared to commercial planes, private jets are 5 to 14 times more polluting and just one hour of private jet flight can emit 2 tons of CO2 [3]. For comparison, the average European citizen emits 8.2 tons of CO2 over the course of an entire year.

Climate change is everyone's burden and responsibility, and it should be addressed at every level, particularly when means and alternatives are available.

1.2 PRIVATE AVIATION

Private aviation has evolved beyond a simple mode of transportation, transcending its utilitarian purpose to embody a symbol of prestige and efficiency. From high-profile individuals to corporate entities, private aircraft offer unparalleled flexibility and convenience, enabling fast travel across vast distances with a level of luxury unmatched by commercial alternatives. However, all these benefits come with one major drawback : its significant contribution to carbon dioxide (CO2) emissions. As the global community grapples with the urgent imperative of combating climate change, the environmental footprint of private aviation has come under increasing scrutiny. This scrutiny is particularly pronounced in the context of European countries, where a mosaic of socio-economic factors, demography, and technological innovations intersect to shape the landscape of CO2 emissions in private aviation. Thus, understanding the multifaceted dynamics driving CO2 emissions disparities across European nations in private aviation is not only imperative for mitigating environmental impact but also for fostering sustainable practices within the aviation industry.

1.3 DATASET

Surprisingly, despite vast quantities of publicly available flight-tracking data, finding quality data was a challenge as most data is not aggregated and includes the entire aviation industry, not just the private aviation sector.

After evaluating our options, we made the decision to base our analysis on data from CE Delft's study, "CO2 emissions from private aviation in Europe" [4], commissioned by Greenpeace. It details the CO2 emissions from private aviation across Europe, emphasizing significant trends and impacts.

CE Delft have aggregated the dataset from flight information provided by Cirium[1] and calculated estimated CO2 emissions with Eurocontrol's Small Emitters Tool [5].

A number of reasonable assumptions were made to create this dataset:

First of all, all planes with less than 3 seats, that landed at or departed from an airport with an IATA code airport and flights that departed from and landed at the same airport were removed from the dataset to exclude leisure and training flights.

However medical and military flights were not filtered out as they cannot be distinguished from private aviation flights. When these flights represented a disproportionate percentage of flights, the authors adjusted the results to have a similar share for each country.

Moreover, the data only calculates CO2 emissions produced during the flight. It does not investigate other types greenhouse gases (such as NOx, SO2) that planes emit. Neither does it take into account indirect CO2 emissions due to production. [6]

By filtering and aggregating this information, the authors included 572 806 flights in their dataset for 2022 and calculated total CO2 emissions to be 3 385 538 tons.

[1] https://www.cirium.com/

CHAPTER 2

IPAT AND KAYA DECOMPOSITION

2.1 WHAT IS AN IPAT AND A KAYA DECOMPOSITION ?

The IPAT equation is an identity that breaks down environmental impact (I) into three key factors: population (P), affluence (A), and technology (T). We can express this simply as $I = P \times A \times T$. This formula highlights how changes in each factor can influence the overall impact we have on the environment.

A useful by-product of the IPAT equation is the Kaya identity. Unlike the IPAT, the Kaya identity specifically focuses on greenhouse gas emissions (and often CO2). It breaks down total emissions into four key factors: population (P), GDP per capita, energy intensity of the GPD (E/GDP) and emission intensity (I/E). The Kaya identity's more fine-grained approach allows for greater precision and a detailed analysis of emissions.

It is always important to keep in mind however that both the IPAT and the Kaya identities have considerable limitations:

First and foremost, it is essential to remember that an IPAT decomposition is merely an identity. It does not ensure that the equation has any meaning if the selected factors are irrelevant to the evaluated impact.

Furthermore, for the IPAT to work, we need the assumption that the components are independent, which means that we risk neglecting potential dependencies and interactions between them that can significantly alter how each factor affects the environmental impact we measure.

Additionally, the IPAT model does not take into account spurious causality which can severly impact findings. The impact can feedback on the population, affluence and technology factors and therefore falsify results.

Finally, these equations often provide an overview using an aggregate population, it can therefore hide severe disparities amongst the population.

To conclude, the IPAT and the Kaya decomposition are extremely intuitive and useful tools to explore the different factors affecting the impact, however they can lead to a limited and potentially deceitful view of the problem.

2.2 KAYA DECOMPOSITION FOR PRIVATE AVIATION

To analyse CO2 emissions of private aviation we use a variation of the Kaya decomposition :

$$CO_2\ emissions = population \cdot \frac{flights}{population} \cdot \frac{kilometers}{flights} \cdot \frac{CO_2\ emissions}{kilometers}$$

We break down CO2 emissions produced by private aviation into four factors :

- **population by country**
- **flights per capita** which indicates how prevalent private flying is among the population and can be compared to the GDP/capita term in the "standard" Kaya decomposition
- **kilometers per flights** which is a parallel to the energy intensity of the Kaya decomposition
- **CO2 emissions per kilometer** which can be seen as the emission intensity

2.3 LIMITATIONS

Our IPAT decomposition has some limitations that need to be addressed, particularly given the specific dataset at hand.

Firstly, the dataset spans only three years: 2020, 2021, and 2022. This limited timeframe may not capture long-term trends or variations in private jet usage and emissions. Additionally, the years 2020 and 2021 were marked by the global COVID-19 pandemic, significantly disrupting air travel patterns. As a result, the data from these years likely underestimate the true CO2 impact of private jets, as air traffic was severely curtailed during this period. Conversely, 2022 saw a rebound in air travel as restrictions eased, potentially leading to a surge in emissions compared to the previous years.

Moreover, the dataset focuses on only the three most flown routes per country. While this approach provides insights into the busiest air corridors, it may not fully capture the diversity of private jet travel patterns within each country. Private jet usage can vary widely depending on factors such as socioeconomic status, geographic location, and business activities, which may not be adequately represented by a limited number of routes, even though it gives an idea of the trends.

Furthermore, the global population of a country, which is used as a proxy for calculating emissions per capita, may not accurately reflect the demographic profile of private jet users. Private jet travel is typically associated with affluent individuals and corporate executives, who constitute a small fraction of the total population but account for a disproportionate share of emissions. Therefore, using overall population figures may lead to an underestimation of per capita emissions from private jet travel.

Taking into account all these limitations of our dataset, we decided, to have the most relevant IPAT analysis possible and to be able to best answer our research question, to focus only on the year 2022 which best represents the current trend because not impacted by COVID-19. In addition, we chose to carry out an analysis by country and another one by city, in order to have more data than the 3 most flown routes because by doing so, we can take all the city-pairs in the dataset in relation to the country, respectively the city.

In conclusion, recognizing these limitations is crucial for interpreting the results accurately. That is why the analyzes which will follow in chapters 3 and 4 were carried out only for the year 2022 and with more data than the 3 most flown routes, which make it possible to compare in the least biased way possible the different factors which contribute to CO2 emissions between European countries.

2.4 UNOBSERVED FACTORS

As mentioned above, the IPAT equation is a powerful tool for analyzing the primary factors contributing to CO2 emissions; however, it omits key unobserved factors that also drive emissions, limiting its overall accuracy.

These unobserved variables could be examined by contrasting differences between countries to see how they might account for variations in the resulting IPAT. However, they are not incorporated into our IPAT and must be analyzed independently.

We will now examine some of these potential unobserved variables :

A significant unobserved variable in our IPAT equation is the effective population utilising these services. Due to data limitations, we base our analysis on the entire population of a country, the majority of whom do not have access to these services. Indeed, the median wealth of a private jet owner is $190 million[7], and the number of these ultra high net worth individuals (UHNWI) varies greatly across countries. As a result, we cannot compare amongst countries the effects of wealth distribution on private jet emissions.

Related to this is the influence of a country's economic structure. For instance, countries with economies heavily centered around finance tend to have more private jet flights compared to those centered around manufacturing sectors [8]. Additionally, the importance of a country's tourism industry also greatly affects the number of private flights and resulting CO2 emissions.

Furthermore, regulatory measures have the potential to significantly impact private jet emissions. For instance, France's ban on short flights when train alternatives under 2h30 exist[9] could notably reduce CO2 emissions, given that 39% of private flights are short-haul (under 500 km)[4].

Another potential unobserved driver of CO2 emissions from private jets is the emergence of new business models (such as fractional ownership and private jet chartering websites and companies) that have made access to these services cheaper and easier[8]. It is hard to quantify such trends as little to no data is available on the subject however the accessibility of private travel through these new services lowers the barrier to entry, potentially leading to a rise in flights in countries where these services have emerged.

Finally, the "technology" factor encompasses multiple elements that cannot be untangled in our IPAT equation. The emission intensity can be affected by various elements such as fuel efficiency of planes, plane size and the use of SAF (sustainable aviation fuels) which can lower emissions by 60%. However, given that SAF only represented 0.1% of total aviation fuel purchase in 2021[7], it is reasonable to assume that the main influences on emission intensity are fuel efficiency and plane size.

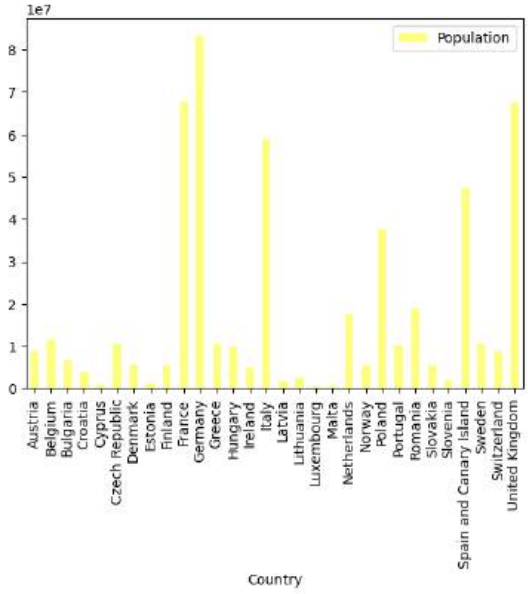
CHAPTER 3

ANALYSIS BETWEEN EUROPEAN COUNTRIES

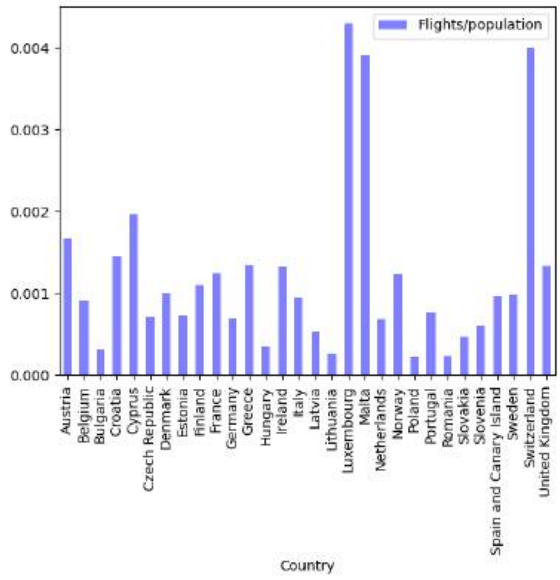
We compared the IPAT decomposition of each European country for the year 2022 to understand what are the factors that contribute to CO2 emissions in the private aviation sector. We will look at how these factors differ across countries and how they can be explained.

For the IPAT decomposition, we had data for each country about the total population, the total number of private jet flights and the total CO2 emissions of this sector. The only missing data point was the total distance travelled by private jets. The top 3 most flown routes for each country were however available. Hence, we calculated the distance for each of these routes, we took the mean number of km per flight and multiplied it by the total number of flights, assuming that the top 3 most flown routes mean distance was approximately representative of the mean total distance.

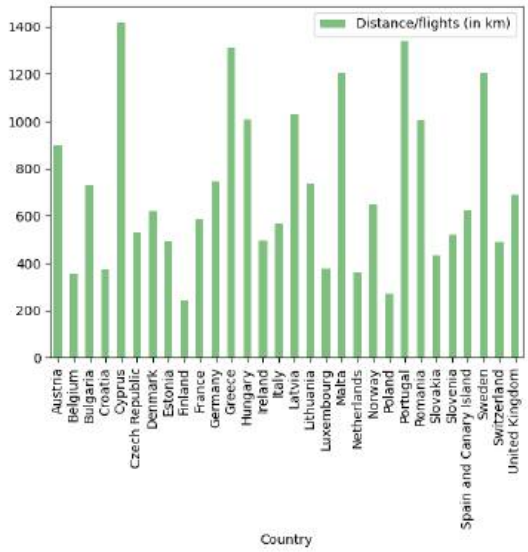
The graphs of each IPAT component can be found on the next page.



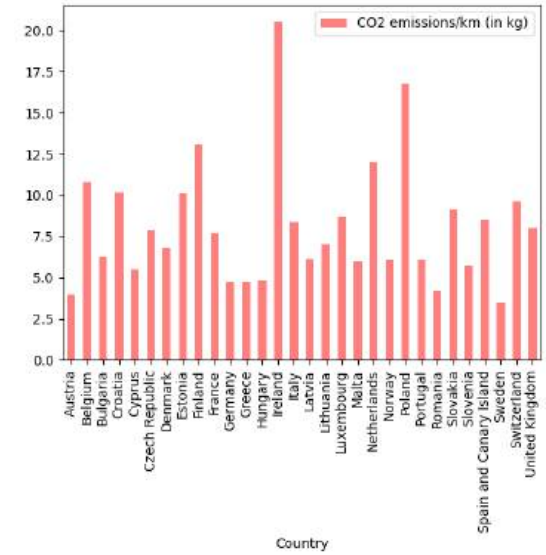
(a) This plot represents the population.



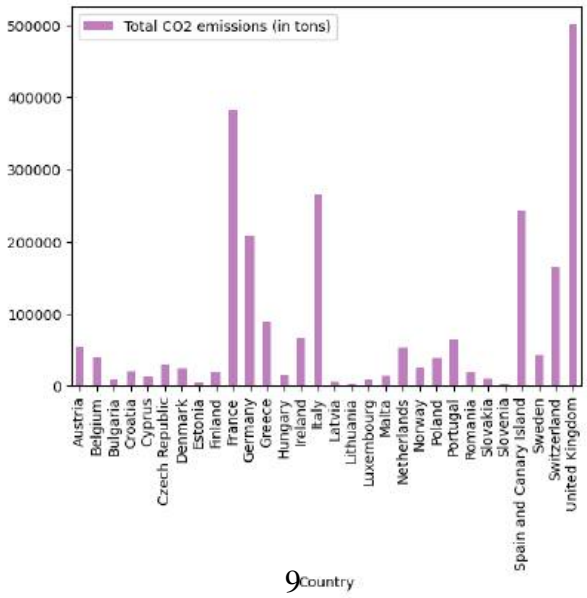
(b) This plot represents the flights per population.



(c) This plot represents the distance per flight.



(d) This plot represents the CO2 emissions per km.



(e) This plot represents the total CO2 emissions.

Figure 3.1: The 5 components of the IPAT decomposition for each european country.

3.1 OBSERVATIONS AND COMPARISONS BETWEEN COUNTRIES

As a first observation, we see that the population and the number of flights per capita are obviously the two biggest drivers of CO2 emissions from private jets.

Furthermore, we can clearly see that there are three countries that use substantially more private jets: Luxembourg, Malta and Switzerland. The three countries have almost four times more flights per population compared to the European average. Luxembourg and Switzerland are the two wealthiest european countries with a strong financial sector, which can be an explanation. For Malta however, the high number of private jet flights is probably more related to its favourable tax regime. The island allows non-domiciled individuals to reduce their tax on aviation income to 0%, which attracts many private jet firms and as a result makes Malta an important hub for the private aviation industry [10].

Then, as expected, the countries at the extremities of Europe travel the further for each flight on average (Cyprus, Greece and Portugal).

We can also see that Ireland, Poland and Finland have the most CO2 intense flights (almost twice kg of CO2 emitted per km compared to the European average). This observation is not completely accurate due to the assumption we made that the top 3 most flown routes distance represent the distance of all the flights. However, by overlaying the two graphs of the average distance per flight and the average CO2 emissions per km, we can clearly see the trend that shorter flights are more CO2 intense Figure 3.2. The other variations in CO2 intensity can be explained by our assumptions and by the type of private jets that are used. This indicates that regulations should first concentrate on short flights due to their high CO2 intensity and the availability of alternative transportation modes that offer comparable convenience.

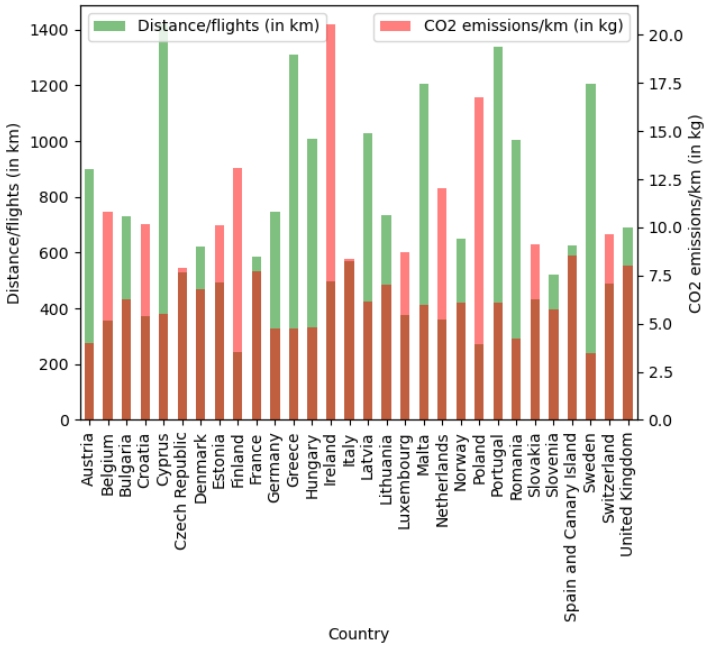


Figure 3.2: The distance per flight on the left axis and the CO2 emissions per km on the right axis, highlighting the higher CO2 intensity of short flights.

Finally, the last graph indicates that the United Kingdom and France emit the most CO2 in total with their private aviation sector.

3.2 CASE STUDY: FRANCE, UK AND GERMANY

When looking at the different IPAT components, it is interesting to see that France and the UK are similar. They almost have the same population, average distance per flight and CO2 emissions per km. The UK only has a slightly higher number of flights per capita which leads to 25% more CO2 emissions. However, when looking at Germany, we notice that it emits less than half CO2 compared to France and the UK, despite having a bigger population and a similar economic prosperity.

This difference in CO2 emissions is mainly due to the relatively low number of flights per capita in Germany. One explanation for this is the difference between the industries present in Germany, France and the UK. According to the Jet Traveler Report from VistaJet [8], the top industries associated to private jet users are:

- For private jet owners: Finance (19%), Real Estate (7%) and Non-Profit & Social Organisations (6%)
- For members of private aviation programs: Finance (21%), Industrial Conglomerate (7%) and Technology (6%)
- For on-demand private fliers: Finance (14%), Manufacturing (8%) and Technology (7%)

Finance which is the number one industry associated to private jets plays a bigger role in the UK's economy compared to France and Germany. But France has also the advantage of being a major tourism hub contributing to a higher private aviation services demand, which is less the case in Germany. Moreover, according to the Globalization and World Cities Research Network (GaWC), London is one of the two cities considered as Alpha++ (most integrated with the global economy) and Paris is considered as Alpha+ (highly integrated cities with the global economy), while Germany doesn't have any city in these two categories which could explain the smaller demand in private jets [11].

CHAPTER 4

CITY ANALYSIS: LONDON, GENEVA, PARIS, AND NICE

In this chapter, we compare the IPAT decomposition of four specific cities: **London**, **Geneva**, **Paris**, and **Nice** for the year 2022. This city-level analysis helps us understand the factors contributing to CO2 emissions in the private aviation sector.

The data for each city were aggregated based on the identification of the city in the "City Pair" column of the 2022 Most Flown Routes dataset for each country [4]. For example, for London, if the *"City Pair"* column was *"London-Dublin"*, we considered the Km, Flight, and CO2 metrics in the total. It's important to note that we didn't take into account the order of the city pair: *"London-Dublin"* or *"Dublin-London"* were treated equally. Population data for the cities are sourced from the *macrotrends* website[1].

The aggregated results are displayed below in table format in Table 4.1 and visually in Figure 5.1. In the *Appendix*, the full aggregated data for the years 2021, 2022, and 2023 for each city can be found, along with some further analysis (2020-2022% growth rate using the yearly mean vs. the between extremes method).

Metric	**London**	**Geneva**	**Paris**	**Nice**
Population	9'541'000	627'000	11'142'000	945'000
Km	115'174'032	18'496'109	14'243'882	32'256'401
Flights	26'379	10'295	13'859	9'657
CO2 (tons)	**110'200**	**30'308**	**37'279**	**36'192**
Flights/Population	0.00276	0.0164	0.00124	0.0102
Km/Flights	4'366.13	1'796.61	1'027.77	3'340.21
CO2/Km	0.0010	0.0016	0.0026	0.0011

Table 4.1: Comparison of 2022 IPAT Factors Across Four Cities

[1] `https://www.macrotrends.net/`

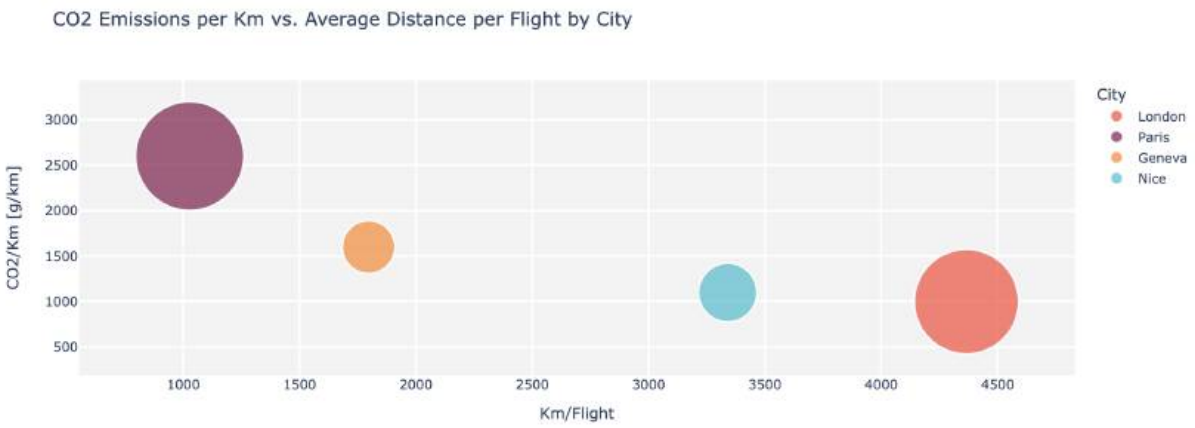


Figure 4.1: This plot represents the CO2 emissions per kilometer [g/km] versus the average distance flown per flight [km/flight] for the four different cities. Each bubble's size corresponds to the city's population, with larger bubbles indicating larger populations.

4.1 OBSERVATIONS AND COMPARISONS BETWEEN CITIES

In examining the factors that contribute to CO2 emissions differences in the private aviation in Table 4.1, **London** stands out with the highest kilometers flown and the highest CO2 emissions. However, **Paris**, with fewer kilometers flown, exhibits the highest CO2 emissions per kilometer. This inefficiency indicates that **Paris** might have a higher proportion of less fuel-efficient, short flights, which typically emit more CO2 per kilometer compared to longer flights.

Geneva, despite having a significantly smaller population, shows a high number of flights per capita, suggesting a strong culture of private aviation, potentially linked to its global role in finance and diplomacy. This is reflected in its high CO2 emissions per kilometer as shown in Figure 5.1, indicating that these flights are potentially less efficient or cover shorter distances where flights are less efficient overall. In contrast, **London** and **Paris**, with their larger populations and more significant economic roles, demonstrate higher overall CO2 emissions but lower emissions per capita due to their spread of private aviation use across a broader base.

4.2 CORRELATION ANALYSIS

Correlation analysis is a powerful tool to explore potential relationships between these metrics. By identifying which factors are closely related, we can pinpoint key drivers of CO2 emissions and target them for efficiency improvements. For this, we will use two correlation metrics.

First, the *Pearson correlation coefficient* is a measure of the linear correlation between two variables, giving a value between -1 and +1. A value of +1 implies a perfect positive linear relationship, -1 implies a perfect negative linear relationship, and 0 implies no linear correlation. This coefficient helps us understand the direction and strength of a linear relationship between two variables. However, the Pearson correlation has its limitations: it only measures linear relationships and might miss out on more complex interactions and can be overly influenced by

outliers, which can distort the perception of a relationship.

Second, the *p-value* in correlation analysis assesses the probability that the observed data would occur if no correlation actually existed. A low *p-value* (typically less than 0.05) indicates that the correlation is statistically significant and not just a result of sampling variability.

Factor 1	Factor 2	Pearson Correlation	P-Value
Km/Flights	CO2/Km	**-0.908328**	**0.091672**
CO2	Km/Flights	0.768552	0.231448
CO2	Flights/Population	-0.525864	0.474136
CO2	CO2/Km	-0.499315	0.500685
Flights/Population	CO2/Km	-0.307952	0.692048
Flights/Population	Km/Flights	-0.116792	0.883208

Table 4.2: Correlation Analysis of the IPAT Factors

As displayed in Table 4.2, the strong negative correlation of **-0.908328** between Km/Flights and CO2/Km indicates that these metrics move in opposite directions. For example, if the distance traveled per flight increases, the amount of CO2 emitted per kilometer tends to decrease, and this could suggest efficiency improvements in flights over longer distances. However, the p-value of **0.091672** tells us that this observed correlation could still reasonably occur if there were no actual underlying relationship in the broader population (i.e., this might be an artifact of the limited sample size).

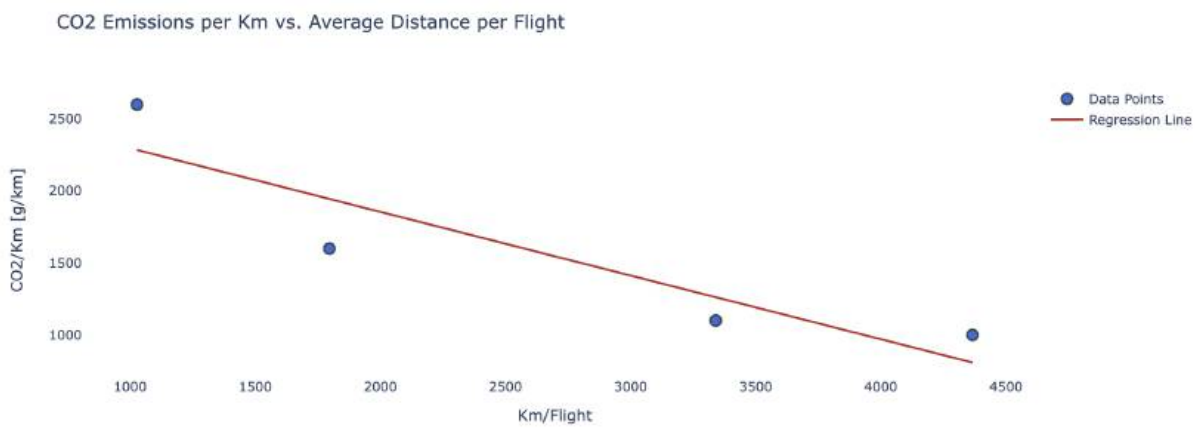


Figure 4.2: This plot represents the CO2 emissions per kilometer [g/km] versus the average distance flown per flight [km/flight] for the cities data. A regression line is added to show the trend and assess the linear relationship between these variables.

For other pairs like (CO2, Km/Flights) and (CO2, Flights/Population), you can see the coefficients are different in magnitude, showing varying strengths of relationships, but their higher *p-values* indicate that these correlations are also not statistically significant.

CHAPTER 5

DIFFERENT WAYS OF MEETING EUROPE NET-ZERO 2050 TARGET

As outlined by the European Commission for Energy, Climate Change & Environment, the European Union has set a climate target to reduce net emissions by 90% by 2040. Furthermore, in its long-term strategy, the EU strives to achieve climate neutrality by 2050, meaning an economy with net-zero greenhouse gas emissions [12]. Europe's Net-Zero 2050 target stands as a monumental commitment in the global fight against climate change. Written in the European Green Deal and reinforced by the European Climate Law, this ambitious goal sets a clear pathway for the European Union to become the world's first climate-neutral continent by 2050.

In the quest to achieve climate neutrality by 2050, the European Union is leading a diverse and comprehensive strategy to tackle emissions across various sectors, including transportation. This sector, known for its significant contribution to greenhouse gas emissions, is witnessing a shift towards electrification and decarbonization. While efforts primarily focus on road transportation, addressing emissions from air transport, including private aviation, is also critical.

5.1 REDUCING IPAT RATIOS

In order to reduce these emissions, it is relevant to look at which factor of the IPAT could be changed. The Population is obviously not a good element to limit. Flights/Population could be restricted through laws and/or heavy taxes, especially for the main private jets users : the UHNWI. Distance/flight is a complicated element because on the one hand it would have to be limited to reduce its impact in the IPAT, but on the other hand, as seen in part 4.2, distance/flight seems to be inversely correlated with CO2 /km. This implies that a short flight with a private jet will have more proportional impact than a long flight. This is why it is not necessarily interesting to reduce the distance per flight but it is more interesting to look at the technology term : CO2/Km. Adopting sustainable aviation fuels (SAFs), investing in research and development of electric and hybrid aircraft, improving aerodynamics on aircraft, using lightweight materials and developing more efficient engines, are all measures that make it possible to drastically reduce CO2 emissions per kilometer. Additionally, optimizing air traffic management systems can further contribute to emissions reduction.

5.2 OTHER TRANSPORT ALTERNATIVES

Although private jets offer certain undeniable advantages for its users, such as flexibility, speed, comfort, etc., it is important to emphasize that in terms of emissions per passenger and per kilometer, its environmental impact is the highest of common means of transport, as shown below [13].

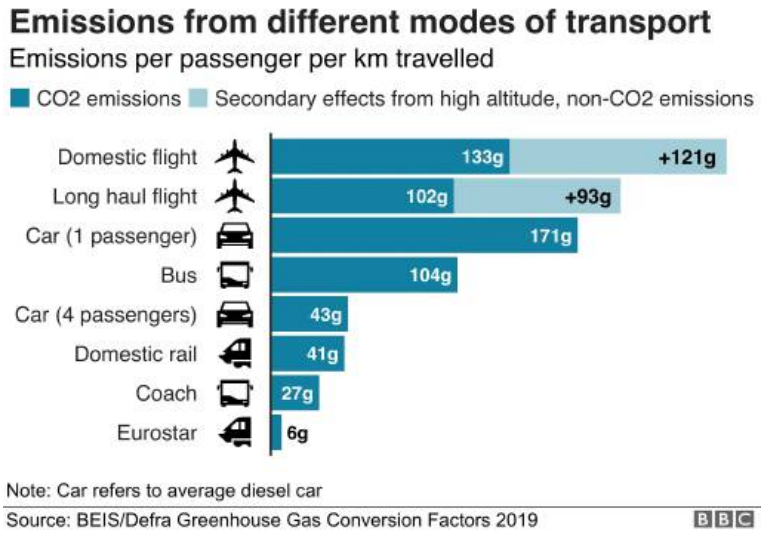


Figure 5.1: This figure from the BBC shows the emissions per passenger per km travelled for different means of transportation (private jets are represented in domestic flight)

It is obviously complicated to have a level of comfort, flexibility and speed equal to that of private jets with the coach or the train. However, according to the VistaJet report, knowing that most private flights are carried out for business or leisure [8], the number of private jet flights could be limited as much as possible. For business, tools like Zoom could be favored or traveling by train in 1st class, where possible, to maintain a certain level of comfort. And as for leisure, leisure activities close to home should be favored, and taking private jets out of habit should be avoided, but only when it is really necessary. Here also, laws and legislation could therefore play a major role in reducing this impact.

5.3 CARBON COMPENSATION

Carbon compensation for private jets might seem like a quick fix to offset their CO2 emissions, but it's not the most effective solution. The main point is to determine the priority destinations for directing carbon offset funds. Redirecting carbon compensation efforts from private jets to more critical sectors, such as agriculture, forestry, or renewable energy projects, can yield far-reaching environmental benefits. These sectors not only emit huge amounts of carbon but also play vital roles in food security, biodiversity conservation, and sustainable development. That is why focusing carbon offset initiatives on sectors with higher emissions and greater potential for positive environmental impact ensures that resources are allocated where they're most needed. Given that private jets are luxury items primarily used for business or leisure, and are not essential for general well-being, it wouldn't be wise to allocate carbon compensation funds toward them.

CHAPTER 6

CONCLUSION

In this report, we have examined the significant impact of private aviation on CO2 emissions within Europe. Through an IPAT and Kaya decomposition analysis, we identified key drivers of emissions in private aviation, highlighting the disparities between European countries and cities. Our findings indicate that the most substantial contributors to CO2 emissions from private jets are the number of flights per capita and the distance traveled per flight. Notably, countries like Luxembourg, Malta, and Switzerland, along with cities such as London, Geneva, Paris, and Nice, exhibit higher emissions due to their unique socio-economic and geographic factors.

To align with the EU's ambitious goal of becoming carbon neutral by 2050, we have explored several strategies to mitigate the environmental impact of private aviation. These include implementing stricter regulations and taxes on private jet usage, especially targeting high-frequency users, and promoting technological advancements to reduce CO2 emissions per kilometer through sustainable aviation fuels, electric and hybrid aircraft, and other innovations. Additionally, shifting travel preferences towards more sustainable modes of transportation and improving air traffic management systems can contribute significantly to emissions reduction.

While carbon compensation schemes offer a way to offset emissions, the primary focus should be on reducing emissions at the source and investing in sectors with higher environmental impact potential, such as renewable energy and sustainable agriculture.

In conclusion, achieving the EU's net-zero target by 2050 requires a multifaceted approach, combining regulatory measures, technological innovation, and behavioral changes. By addressing the disproportionate impact of private aviation and encouraging sustainable practices, we can move closer to a climate-neutral future, ensuring that the benefits of progress do not come at the expense of our planet's health.

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CHAPTER 7

APPENDIX

London	**2020**	**2021**	**2022**	**2020-2022% (mean)**	**2020-2022% (extremes)**
Population	9’304’000	9’426’000	9’541’000	1.27%	2.55%
Km	1’529’238	5’866’213	115’174’032	1073.47%	7431.47%
Flights	2’427	8’024	26’379	229.68%	986.90%
CO2 (tons)	6’815	29’914	110’200	303.67%	1517.02%
Flights/Population	0.00026	0.00085	0.00276	225.56%	959.90%
Km/Flights	630.09	731.08	4366.13	256.62%	592.93%
CO2/Km	0.0045	0.0051	0.0010	-33.41%	-78.53%

Table 7.1: Statistical Data Overview of London from 2020 to 2022

Paris	**2020**	**2021**	**2022**	**2020-2022% (mean)**	**2020-2022% (extremes)**
Population	11’017’000	11’079’000	11’142’000	0.57%	1.13%
Km	867’272	4’217’227	14’243’882	312.01%	1542.38%
Flights	2’320	9’584	13’859	178.85%	497.37%
CO2 (tons)	4’450	25’489	37’279	259.52%	737.73%
Flights/Population	0.0002	0.0009	0.0012	177.29%	490.67%
Km/Flights	373.82	440.03	1027.77	75.64%	174.93%
CO2/Km	0.0051	0.0060	0.0026	-19.45%	-48.99%

Table 7.2: Statistical Data Overview of Paris from 2020 to 2022

Geneva	**2020**	**2021**	**2022**	**2020-2022% (mean)**	**2020-2022% (extremes)**
Population	613'000	620'000	627'000	1.14%	2.28%
Km	1'210'798	2'808'709	18'496'109	345.25%	1427.60%
Flights	2'691	6'303	10'295	98.78%	282.57%
CO2 (tons)	5'510	16'217	30'308	140.60%	450.05%
Flights/Population	0.0044	0.0102	0.0164	96.55%	274.03%
Km/Flights	449.94	445.61	1'796.61	151.11%	299.30%
CO2/Km	0.0046	0.0058	0.0016	-22.37%	-63.99%

Table 7.3: Statistical Data Overview of Geneva from 2020 to 2022

Nice	**2020**	**2021**	**2022**	**2020-2022% (mean)**	**2020-2022% (extremes)**
Population	942'000	943'000	945'000	0.16%	0.32%
Km	314'534	3'267'762	32'256'401	913.02%	10155.30%
Flights	899	4'337	9'657	252.55%	974.19%
CO2 (tons)	1'621	16'986	36'192	530.47%	2132.70%
Flights/Population	0.0010	0.0046	0.0102	252.05%	970.78%
Km/Flights	349.87	753.46	3'340.21	229.33%	854.70%
CO2/Km	0.0052	0.0052	0.0011	-38.78%	-78.23%

Table 7.4: Statistical Data Overview of Nice from 2020 to 2022